

Synheart Kinematic RulePack

Mathematical documentation of the deterministic kinematic axes

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Chapter 1

Activity State

Activity State estimates *the type and intensity of whole-body movement* — sedentary, walking, running, or cycling — from the strength and frequency of acceleration. It is theoretically distinct from **Postural State** (which measures body orientation, not whether or how the body is moving) and from **Movement Regularity** (which measures how rhythmic a motion is, not what kind it is or how energetic). This is the foundational axis: it labels *what the body is doing*, and the other axes either build on that label or resolve the questions it deliberately leaves open.

Scope. This axis reports the dominant activity *class* over a short window, not a continuous intensity. Standing folds into *sedentary* on purpose — posture is a separate question, delegated to Postural State — and “vigorous” is not modelled as a class above running. The confound most likely to occur is **walking versus running**: their energy distributions genuinely overlap, and the split in that overlap rests on cadence and on placement-specific thresholds. Cycling can only be detected when a gyroscope is present; without one it reads as sedentary (pedalling is low in linear acceleration).

1.1 Theoretical Basis

The rule is an *accelerometer intensity cut-point* classifier, the decades-old method of the physical-activity-monitoring literature, extended with a cadence tie-breaker and a rotational cue for cycling. The core feature — the standard deviation of the acceleration magnitude — is the same construct as **MAD** (Mean Amplitude Deviation, Vähä-Ypää et al., 2015) and **ENMO** (Euclidean Norm Minus One, Hildebrand et al., 2014, the GGIR default), and the cut-point idea traces to activity-count thresholds (Freedson et al., 1998; Troiano et al., 2008). What is *directly evidenced* by that work is the use of a magnitude-intensity statistic against fixed thresholds. The *engineering adaptations*, flagged below, are the specific threshold values (re-derived on MotionSense), the cadence tie-breaker inside the overlap band, and the gyroscope cycling branch.

- **Acceleration magnitude intensity** — $\text{std}(|a_{\text{body}}|)$ separates still from moving and low- from high-energy gait; it is orientation-invariant, so it does not depend on how the device is carried (Yurtman & Barshan, 2017).
- **Cadence (step frequency)** — where energy alone cannot separate a fast walk from a slow run, step rate can; walking and running occupy different cadence ranges in the gait literature.
- **Rotational energy for cycling** — pedalling produces little linear acceleration but strong limb/segment rotation, so mean gyroscope magnitude detects cycling that the accelerometer misses.

1.2 Input Signals

Signal	Symbol	Description
Body acceleration	a_{body}	Gravity-removed 3-axis acceleration (m/s ²)
Motion size	move	$\text{std}(a_{\text{body}} /g)$ over the window (g)
Energy	energy	$ a_{\text{body}} /g$, mean magnitude (g)
Cadence	cad	Step frequency (Hz) from the autocorrelation peak
Gyroscope energy	gyro	$ \omega $, mean angular-rate magnitude (rad/s); cycling cue

Window 128 samples (2.56 s at $F_s = 50$ Hz), 50% overlap — the shortest of the axes, so activity reacts quickly. No personal baseline: the thresholds are population-derived and frozen.

1.3 Rule Formula

RULE-AS *Activity State Classifier*

Step 1 — Features ($s = |a_{\text{body}}|/g$; cadence over lag band [10, 64]):

$$\text{move} = \text{std}(s), \quad \text{energy} = \bar{s}, \quad \text{cad} = \frac{F_s}{k^*}, \quad \text{gyro} = |\omega|.$$

Step 2 — Still gate:

$$\text{move} < 0.10 \text{ g} \Rightarrow \text{sedentary}.$$

Step 3 — Walk / run split (else, i.e. moving):

$$\begin{aligned} 0.77 \leq \text{energy} \leq 1.07 : & \quad \text{running if } \text{cad} \geq 1.20 \text{ Hz, else walking} \\ \text{otherwise} : & \quad \text{running if } \text{energy} \geq 1.00 \text{ g, else walking} \end{aligned}$$

Step 4 — Cycling override (only in the still region, needs gyro):

$$(\text{move} < 0.10 \wedge \text{gyro} > 0.18 \text{ rad/s}) \Rightarrow \text{cycling}.$$

$$\text{label} \in \{\text{sedentary, walking, running, cycling}\}$$

Example (real window, § 1.8): **move** = 0.54, **energy** = 0.91 (in the overlap band $\Rightarrow$ energy cannot decide), **cad** = 1.43 $\geq$ 1.20 $\Rightarrow$ **running**.

1.4 Parameter Notes

Still threshold $T_{\text{STILL}} = 0.10 \text{ g}$. Below this, magnitude variation is sensor noise, not body motion. Re-derived on MotionSense; the UCI-HAR value (0.05) did *not* transfer because of the unit/placement difference, which is why every threshold here is stated in g on the same body-acceleration scale.

Run energy line $T_{\text{RUN}} = 1.00 \text{ g}$ **and overlap band** [0.77, 1.07]. Outside the band, mean magnitude alone separates walking from running. Inside the band the two classes' energy distributions touch (train-supported edges), so energy is *not* trusted there — the cadence tie-breaker takes over.

Cadence split $CAD_{\text{RUN}} = 1.20 \text{ Hz}$ (**provisional**). Used *only* inside the overlap band. It is placement-specific: on waist data (HHAR) the autocorrelation fundamental flips between step and stride, so this value does not transfer unchanged and is flagged provisional pending own-phone pocket data.

Gyro cycling threshold $T_{\text{GYRO}} = 0.18 \text{ rad/s}$ (**provisional**). Derived on HHAR (waist); pocket cycling is not yet verified. The branch fires only in the still region so it cannot corrupt walking/running.

Fusion method. This is a *priority decision cascade*, not a weighted sum. Energy decides first; cadence is a *tie-breaker* used only where energy is ambiguous; the gyroscope is an *override gate* confined to the still region. The ordering encodes which cue is trusted in which regime, keeping each class’s decision traceable to one deciding quantity.

Provenance — where each number comes from

Every constant is traceable to a dataset or a stated convention; none is a free guess. The labels mean: DATA = computed from the data by a formula; FITTED = read off the observed feature distributions and then validated on held-out subjects; CONV = a standard/physiological choice, not tuned; PROV = data-informed but not locked (placement-specific); DESIGN = an engineering construction.

Constant	Value	Data	How it was set	Tag
Window / overlap	128/64	—	UCI-HAR standard 2.56 s window, 50% overlap	CONV
Cadence lag band	[10, 64]	—	Human cadence 0.78–5 Hz at 50 Hz	CONV
T_{STILL}	0.10 g	MotionSense	Read off the still/moving split in the feature plot; held-out validated	FITTED
T_{RUN}	1.00 g	MotionSense	Read off the walk/run energy separation; held-out validated	FITTED
Overlap band	[0.77, 1.07] g	MotionSense	Empirical edges of the walk/run energy overlap	DATA
CAD_{RUN}	1.20 Hz	MotionSense	Walk/run cadence split; did not transfer to waist	PROV
T_{GYRO}	0.18 rad/s	HHAR	Gap midpoint $(\text{sed}_{p95} + \text{cyc}_{p05})/2$	PROV
Confidence proxy	0.5 + 0.5 clamp(m)	—	Threshold-distance construction (live)	DESIGN

The FITTED thresholds (T_{STILL} , T_{RUN}) are round numbers chosen by reading the MotionSense training distributions, not the output of an optimizer; their justification is the held-out accuracy (§ 1.9), not a closed-form derivation. The authoritative derivation lives in the `activity_state.ipynb` notebook.

1.5 Confidence

Confidence is a deterministic *margin proxy* in $[0.5, 1.0]$: how far the deciding quantity sits from the threshold that chose the label.

Step C1 — Deciding margin m (by the branch that fired):

$$m = \begin{cases} (\text{gyro} - 0.18)/0.18 & \text{cycling} \\ (0.10 - \text{move})/0.10 & \text{sedentary} \\ |\text{cad} - 1.20|/1.20 & \text{walk/run inside the band} \\ (\text{energy} - 1.00)/1.00 & \text{running (outside band)} \\ (1.00 - \text{energy})/1.00 & \text{walking (outside band)} \end{cases}$$

Step C2 — Confidence: $\boxed{\text{conf} = 0.5 + 0.5 \text{ clamp}(m, 0, 1)}$

Confidence ceiling / caveat. This is a *proxy*, not the trained, calibrated confidence. On the source bench (MotionSense) the calibrated model reaches ECE = 0.024; the proxy reads 0.5 at a decision boundary (maximum uncertainty) and 1.0 far from it. Off-source it is less calibrated, so it should be read as a relative distance-to-boundary, not a probability.

1.6 Interpretation Scale

Activity State is categorical; the “scale” is the class set and each class’s signature.

Class	Signature	Meaning
Sedentary	$\text{move} < 0.10 \text{ g}$	Still or near-still: sitting, standing, resting (posture delegated)
Walking	Moving, below the run line / in-band cadence $< 1.20 \text{ Hz}$	Steady low-to-moderate gait
Running	$\text{energy} \geq 1.00 \text{ g}$, or in-band cadence $\geq 1.20 \text{ Hz}$	High-energy, fast-cadence gait
Cycling	Still-region body motion with $\text{gyro} > 0.18 \text{ rad/s}$	Low linear motion but rotational (pedalling); needs a gyroscope

1.7 Context Applicability

Target context: a phone or wearable carried on the body (pocket-scale placement), sampling accelerometer and gyroscope at 50 Hz.

Context	accel	gyro	Reliable?
Phone in pocket (target)	Yes	Yes	Yes — all four classes
Waist / no gyroscope	Yes	No	Partial — cycling reads as sedentary
Wrist / other placement	Yes	Partial	Partial — cadence/gyro thresholds shift

Scientific boundary. Even with ideal data the axis cannot separate *standing* from *sitting* or from sedentary — that needs orientation, delegated to Postural State. It does not model

“vigorous” as a class above running. The walking/running boundary is a real energy overlap resolved heuristically by cadence, and both the cadence and gyro thresholds are placement-specific rather than universal.

1.8 Worked Example: Resolving a Jog in the Overlap Band

Scenario (real captured window). A user jogs at a steady pace, phone in the front pocket. The window below is a genuine capture, not an invented one — MotionSense subject 19 (held out from all threshold derivation), jog trial, 58.9s in (`jog_9/sub_19.csv`, samples 2944–3071) — run through the live axis. Its energy lands inside the walk/run overlap band, so energy alone cannot decide the class.

Signal	Value
move (std of s)	0.54 g
energy (mean of s)	0.91 g (in band [0.77, 1.07])
cad	1.43 Hz
gyro	1.87 rad/s

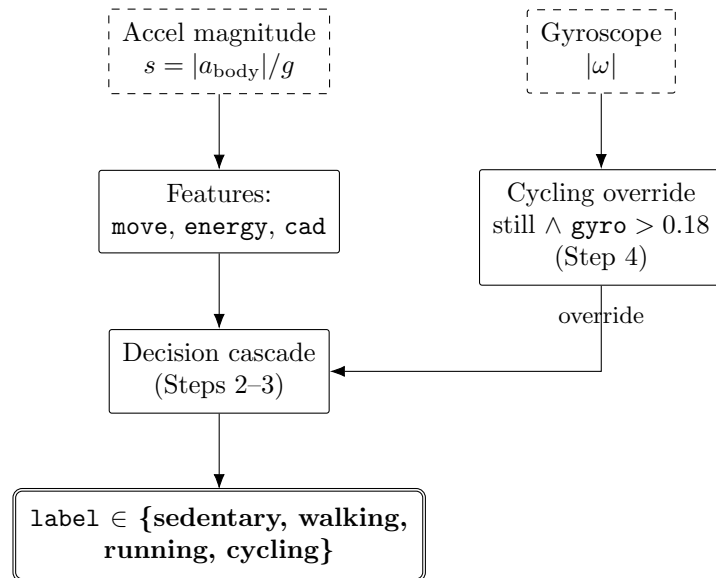
Step-by-step:

$$\begin{aligned}
 \text{move} &= 0.54 \geq 0.10 \Rightarrow \text{not sedentary} \\
 \text{energy} &= 0.91 \in [0.77, 1.07] \Rightarrow \text{use cadence, not energy} \\
 \text{cad} &= 1.43 \geq 1.20 \Rightarrow \mathbf{\text{running}} \\
 \text{move} &< 0.10 \text{ false} \Rightarrow \text{cycling override does not fire} \\
 m &= |1.43 - 1.20|/1.20 = 0.19, \quad \text{conf} = 0.5 + 0.5 \cdot 0.19 = \mathbf{0.60}
 \end{aligned}$$

Interpretation: the window is classified *running*. Because the energy fell in the overlap band, the decisive driver was the *cadence* (1.43 Hz), not the energy. Confidence is moderate (0.60): the step rate is above but not far above the 1.20 Hz walk/run line, which is exactly the region where this axis is least certain. The gyroscope reads high (1.87 rad/s) but is ignored — the cycling override only fires in the still region, and here the body is clearly moving.

1.9 Implementation Notes

- **Window.** 128 samples (2.56 s) at 50 Hz, 50% overlap — the shortest window in the pack, so the activity label updates fastest.
- **Fallback behavior.** No accel/linear stream $\rightarrow$ null. No gyroscope $\rightarrow$ the cycling branch is disabled and the reading note says so; the accel classes are unaffected. Cadence still computes from the accelerometer.
- **Relationship to other chapters.** The standing/sitting ambiguity it folds into sedentary is resolved by **Postural State**; its walking/running output gates **Movement Regularity** (only meaningful while moving) and feeds **Locomotion** and the **Overall State** combiner.
- **Validation status.** MotionSense (subject-independent): 97.1% accuracy, macro-F1 0.948, ECE = 0.024 with the cadence tie-breaker; the only real confusion is walk $\leftrightarrow$ run at the energy overlap. HHAR frozen cross-dataset: shared classes (sedentary, walking) macro-F1 0.996. Cycling via the gyro branch: F1 0.92 in isolation, 0.829 integrated. CAD_{RUN} and T_{GYRO} remain provisional pending own-phone pocket data.



Formula flow — Activity State: a priority decision cascade over orientation-invariant accelerometer features, with a gyroscope override for cycling.

Chapter 2

Movement Regularity

Movement Regularity estimates *how rhythmic and self-repeating the body’s motion is over a short window* — a smoothness score, not an intensity or a posture. It is theoretically distinct from **Activity State** (which measures *how much* motion and of *what type*) and from **Postural State** (which measures orientation relative to gravity). A person can be highly active yet irregular (stumbling, fidgeting) or barely active yet perfectly regular (a slow steady walk); regularity is the axis that separates these, so it is a genuinely non-redundant construct rather than a re-scaling of activity intensity.

Scope. This score reports *whether* motion repeats cleanly, not *why* it does or does not. A low score cannot by itself distinguish a mechanical cause (rough terrain, an obstacle) from a physiological one (tremor, agitation, fatigue). The score is only defined while the body is actually moving rhythmically; when the person is still, or when too few motion cycles fall inside the window, the axis returns **null** rather than a misleadingly “perfect” value. Resolving the cause of irregularity requires the Activity State context and the downstream heart-rate / affective layer.

2.1 Theoretical Basis

The score is the normalized *unbiased autocorrelation* of the acceleration-magnitude signal at its dominant period. This is the trunk-accelerometry gait-regularity index of **Moe-Nilssen & Helbostad (2004)**: for a walking signal, the value of the normalized autocorrelation at the stride lag measures how closely one stride reproduces the next, and the authors note it is cheap enough for portable devices. The parts of the formula *directly evidenced* by that work are the unbiased-autocorrelation estimator and the use of its dominant-period peak height as the regularity index. The remaining choices are *engineering adaptations*, flagged as such below: the clamp to $[0, 1]$, the movement and cycle-count gates, the pocket-placement lag band, and the two-factor confidence.

- **Acceleration magnitude** $|a_{\text{body}}|$ — gravity-removed body acceleration reduced to its Euclidean norm, making the feature orientation-invariant (the same rhythm reads identically regardless of how the phone is rotated in the pocket).
- **Unbiased autocorrelation** — the $1/(N - k)$ normalization corrects the end-of-window sample deficit so that longer lags are not artificially damped (Moe-Nilssen & Helbostad, 2004).
- **Smoothness alternatives (not used, cited for completeness)** — spectral arc length SPARC (Balasubramanian et al., 2012) and log dimensionless jerk (LDLJ) are established

smoothness measures; autocorrelation is chosen here because it is period-aware and matches the gait literature.

- **Affective relevance** — reduced or irregular ambulatory movement is associated with agitation and anxiety in actigraphy studies, which motivates exposing regularity to the downstream affective read.

2.2 Input Signals

Signal	Symbol	Description
Body acceleration	a_{body}	Gravity-removed 3-axis acceleration (m/s^2); linear-accel sensor, or raw accel low-passed at 0.3 Hz
Acceleration magnitude	s	$ a_{\text{body}} /g$, orientation-invariant motion size (g)
Dominant lag	k^*	Period (samples) of the strongest self-repeat in the window
Activity state	ACT	Context gate: regularity is only meaningful while walking/running

Window $N = 250$ samples (5 s at $F_s = 50$ Hz), 50% overlap. No personal baseline is required: the score is self-referential within the window.

2.3 Rule Formula

RULE-MR *Movement Regularity Score*

Step 1 — Magnitude and detrend:

$$s(t) = \frac{|a_{\text{body}}(t)|}{g}, \quad x(t) = s(t) - \bar{s}, \quad \text{movement} = \text{std}(s).$$

Step 2 — Unbiased, normalized autocorrelation (lags $k = 0 \dots K$, $K = 100$):

$$r(k) = \frac{1}{N-k} \sum_{t=0}^{N-1-k} x(t) x(t+k), \quad \rho(k) = \frac{r(k)}{r(0)}.$$

Step 3 — Dominant period (argmax over the stride band, not the first peak):

$$k^* = \arg \max_{k \in [10, 100]} \rho(k), \quad n_{\text{cyc}} = \frac{N}{k^*}.$$

Step 4 — Score:

$$R = \text{clamp}(\rho(k^*), 0, 1).$$

Step 5 — Validity gate (return **null** instead of a score if):

$$\text{movement} < 0.051 \text{ g (low_movement)} \quad \text{or} \quad n_{\text{cyc}} < 3.5 \text{ (insufficient_cycles)}.$$

$$R \in [0, 1] \text{ if valid, else null}$$

Example: steady pocket walk, $\text{std}(s) = 0.37 \text{ g}$, $k^* = 55$ ($\approx 0.9 \text{ Hz}$ stride), $\rho(k^*) = 0.71$, $n_{\text{cyc}} = 250/55 = 4.5$. Both gates pass $\Rightarrow R = 0.71$ (“rhythmic”).

2.4 Parameter Notes

Lag band [10, 100] samples. At 50 Hz this covers stride periods 0.2–2.0 s (0.5–5 Hz), the plausible human cadence range. Taking the *argmax* over the whole band (rather than the first prominence peak) is deliberate: on pocket data the autocorrelation has strong sub-stride harmonics, and the first peak grabs a half-stride and inverts the rank ordering. This is the FIX 2 correction re-derived during the chest→pocket convergence test.

Movement floor 0.051 g. Below this, magnitude variation is sensor noise, not motion, and the autocorrelation of noise produces a spurious high peak. The value is the chest-derived 0.5 m/s^2 floor re-expressed in g ($0.5/9.81$); expressing it in stream units was FIX 1 of the convergence test.

Minimum cycles 3.5 strides. Fewer than ~ 3 –4 repeats in the window makes a “regularity” estimate statistically meaningless. 3.5 stride-periods is the gait-literature adequacy minimum, re-expressed for the argmax-selects-stride convention on pocket placement.

Fusion method. There is *no* multi-signal fusion in the score itself — it is a single autocorrelation read. Fusion appears only in the *confidence* (§ Confidence), which combines two *independent* adequacy factors (enough cycles, and a prominent peak) as a geometric mean, so that either factor being weak pulls confidence down.

Provenance — where each number comes from

Tags: THEORY = taken directly from the published method; CONV = a standard/physiological choice; DESIGN = an engineering construction.

Constant	Value	Data	How it was set	Tag
Regularity estimator	unbiased autocorr. peak	—	Moe-Nilssen & Helbostad (2004) trunk-gait index	THEORY
Lag band	[10, 100]	—	Stride periods 0.2–2.0 s at 50 Hz	CONV
Movement floor	0.051 g	—	Chest 0.5 m/s ² noise floor re-expressed in g (unit fix)	DESIGN
Min. cycles	3.5 strides	—	Gait-literature adequacy minimum (~3–4 strides)	CONV
Prominence sat. (P_{full})	1.5	—	Point at which the confidence term saturates	DESIGN
Confidence	$\sqrt{c_{\text{cyc}} c_{\text{prom}}}$	—	Geometric mean of two adequacy factors	DESIGN

The estimator is the one number taken straight from published theory; the score’s *separation* was validated on PAMAP2/MotionSense (§ 1.9), while the gates and confidence are design choices calibrated to keep noise from scoring as rhythm. The authoritative build lives in the `movement_regularity.py` axis and the convergence-test notebook.

2.5 Confidence

Step C1 — Cycle adequacy (how far past the minimum the window sits):

$$c_{\text{cyc}} = \text{clamp}\left(\frac{n_{\text{cyc}} - 3.5}{5.0 - 3.5}, 0, 1\right).$$

Step C2 — Peak prominence (p = prominence of the peak at k^* ; 1.5 = saturation):

$$c_{\text{prom}} = \text{clamp}\left(\frac{p}{1.5}, 0, 1\right).$$

Step C3 — Final confidence (geometric mean of the two): $\boxed{\text{conf} = \sqrt{c_{\text{cyc}} \cdot c_{\text{prom}}}}$

Confidence ceiling / caveat. This axis reports no labelled accuracy by design — it is validated by *separation* (rank-ordering rhythmic above irregular above still), not against a ground-truth “regularity” label, which does not exist in the datasets. Confidence therefore expresses *internal signal quality* (enough clean cycles), not calibrated correctness, and should not be read as a probability.

2.6 Interpretation Scale

Score	Level	Meaning
0.0–0.2	Erratic	Motion present but not self-repeating (stumbling, fidgeting)
0.2–0.4	Low regularity	Weak, inconsistent rhythm
0.4–0.6	Moderate	Discernible but imperfect rhythm
0.6–0.8	Rhythmic	Clear, steady periodic motion (typical walking)
0.8–1.0	Highly regular	Metronomic, strongly self-repeating (steady running/pace)
<i>null</i>	Not applicable	Still, or too few cycles — no rhythm to score

2.7 Context Applicability

Target context: on-body (pocket / trunk) accelerometry during *sustained locomotion*. The score is most reliable when the Activity State is walking or running.

Context	$ a_{\text{body}} $	$\text{cycles} \geq 3.5$	Reliable?
Walking / running (pocket or trunk)	Yes	Yes	Yes — full
Cycling / non-gait rhythmic motion	Yes	Partial	Partial (period may fall outside band)
Standing / sitting / lying	No	No	No — returns null

Scientific boundary. Even with ideal data, the score cannot attribute *cause*: it measures the presence of rhythm, not its origin, so it approximates “movement smoothness” rather than directly measuring any affective or neuromotor state. Rhythmic-versus-irregular separation on pocket placement is confirmed only for the still-versus-rhythmic contrast; irregular-but-moving on pocket is not yet validated on own-phone data.

2.8 Worked Example: Steady Commute Walk

Scenario. A user walks to a bus stop at a relaxed, even pace, phone in the front trouser pocket. Over one 5 s window the pipeline extracts:

Signal	Value
$\text{std}(s)$ (movement)	0.37 g
Dominant lag k^*	55 samples (≈ 0.9 Hz stride)
$\rho(k^*)$	0.71
Peak prominence p	0.60

Step-by-step:

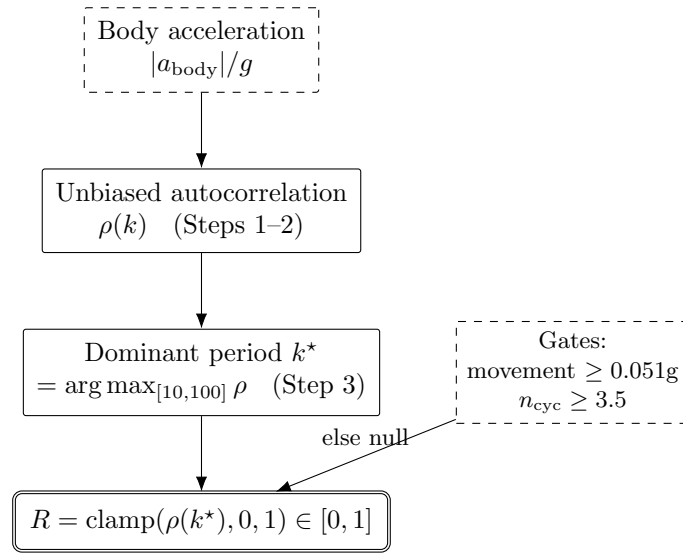
$$\begin{aligned}
 n_{\text{cyc}} &= 250/55 = 4.5 \ (\geq 3.5, \text{ movement } 0.37 > 0.051 \Rightarrow \text{valid}) \\
 R &= \text{clamp}(0.71, 0, 1) = \mathbf{0.71} \\
 c_{\text{cyc}} &= \text{clamp}((4.5 - 3.5)/1.5, 0, 1) = 0.67 \\
 c_{\text{prom}} &= \text{clamp}(0.60/1.5, 0, 1) = 0.40 \\
 \text{conf} &= \sqrt{0.67 \cdot 0.40} = \mathbf{0.52}
 \end{aligned}$$

Interpretation: $R = 0.71$ falls in the *Rhythmic* band — a clean, steady walking cadence. The dominant driver is the high autocorrelation peak ($\rho = 0.71$); confidence is moderate (0.52), held down by the modest peak prominence rather than by too few cycles.

2.9 Implementation Notes

- **Window.** 250 samples (5 s) at 50 Hz, 50% overlap; magnitude is read from the same rolling buffer the other 5 s axes use.

- **Fallback behavior.** No accel/linear stream $\rightarrow$ null. A window that is $> 50\%$ bad samples (NaN or > 16 g), or whose first sample is bad, $\rightarrow$ null (`window_unreliable`); shorter gaps are forward-filled. If `scipy` is unavailable the score still computes; only the prominence confidence factor degrades.
- **Relationship to other chapters.** Gated by **Activity State** — a regularity number is only surfaced while ACT indicates locomotion; on its own, regularity cannot tell moving-irregular from still, which is why the two axes are read together.
- **Validation status.** Continuous Moe-Nilssen score built on PAMAP2 trunk data (rank-separation ROC-AUC ≈ 0.95 ; no labelled accuracy by design). Chest $\rightarrow$ pocket placement transfer of the period selection and movement floor is proven on held-out MotionSense (walking 0.71, running 0.66, both above sedentary ~ 0.30). Still owed to own-phone data: irregular-but-moving separation on pocket, and cycling-cadence handling.



Formula flow — Movement Regularity: one orientation-invariant input, period-aware autocorrelation, a validity gate, and a $[0, 1]$ score.